

When Does PEFT Ensembling Help?

A Pre-Registered Adjudication of 1,028 Cells

Anonymous ACL submission

Abstract

We pre-register and release ADAPTERFRONTIER, a 1,028-cell corpus (registered at git SHA 02a5a6bc *before any training*; 67 adapter pools, 10 architectures, 8 tasks)¹ that adjudicates when PEFT adapter ensembling actually helps versus a compute-matched single adapter. **Ensembling is conditional:** 33.0% of cells are SUPPORTED, 17.8% are REVERSED (ensemble significantly *underperforms* the baseline; 29.5%/16.9% after corpus-wide Benjamini-Hochberg FDR at $q=0.05$) — structure that aggregate task-level reporting destroys. **The boundary is predictable:** a diversity-quality regression attains LOPO-CV $R^2=0.60$ on encoder accuracy (95% CI $[+0.30, +0.71]$), decomposing into five actionable rules (D1–D5). **Following D1–D5** over the field-standard blind `soft_vote` turns encoder $\Delta=-3.8$ pp into $+0.5$ pp at $20\times$ less inference cost; REVERSED rate drops 11%→2% (App. P; oracle-over-corpus, L11). **Mechanism:** within our released feature set, prediction-space disagreement is the dominant out-of-pool predictor (weight-space distance is irrelevant); a 21% lower $H(\text{correctness} | \text{confidence})$ in decoders motivates distillation, which recovers 87% of ensemble gain on Qwen-0.5B MNLI but only 4.5% on Qwen-1.5B MNLI (1-of-2 tested decoder cells) and fails on encoders — not yet a generalizable amortization. **OOD honesty:** on 19 MNLI-trained pools evaluated on ANLI dev_r3, soft-vote accuracy gain is positive on only 5/19 pools (Pool-B HP-diverse: 0/6) — ensembling does not transfer the in-distribution gains to severe distribution shift. **GSM8K honesty:** a digit-shift probe on Qwen-2.5-7B drops accuracy 0.66→0.04, indicating heavy pre-training contamination; our decoder GSM8K gains should be read as "ensemble effects on the

contaminated benchmark" (L12). **Release:** the 67 pools, 1,028 per-cell paired-bootstrap CIs, Holm-corrected verdicts, and full provenance; new ensembling methods adjudicate against released baselines by supplying per-example predictions only — no retraining of the existing 67 pools is required.

1 Introduction

Parameter-efficient fine-tuning (PEFT) of large language models with LoRA-style adapters (Hu et al., 2022) has made it routine for NLP practitioners to keep dozens of task-specific adapters around: serving libraries such as vLLM and S-LoRA (Kwon et al., 2023; Sheng et al., 2024) now route per-request to one of N adapters with negligible base-model duplication, and frameworks such as LoRAHub (Huang et al., 2023) explicitly target multi-adapter composition. A natural question is whether *combining* the available adapters — by averaging weights, voting on labels, or routing requests through them — beats the best single adapter at a fixed compute budget. Recent work reports a wide range of answers: positive on some NLP tasks, negative on others, with no consensus on when ensembling is worth the cost. NLP practitioners currently lack a principled way to decide *a priori* whether to ensemble their adapter pool for a given (task, model) cell.

We argue this disagreement is partly an artifact of *aggregate* reporting. We define the *diversity-quality frontier* as the leave-one-pool-out (LOPO) cross-validated regression $\hat{\Delta}=f(\phi)$ predicting per-cell ensemble gain Δ from a feature vector ϕ of pool-diversity metrics (App. M); R^2_{LOPO} quantifies whether the right-method-for-this-cell is predictable *a priori*. In a 1,028-cell pre-registered, compute-matched adjudication of 67 PEFT pools across 10 architectures and 8 tasks, we find:

- **Decoder GSM8K majority_vote SUPPORTED across $N=6$ cells; two compensat-**

¹Cell count: 1,028 = 827 v1.0 main-pool cells (55 pools $\times$ ensemble methods $\times$ {best_of_n, n_rank} $\times$ {accuracy, ECE}) + 100 v1.1 HellaSwag multichoice cells + 101 baseline self-comparison cells. The 2 v1.1 GSM8K and 7 v1.2 supplementary pools (App. K) are reported separately in Table 3 and are not in the headline 1,028.

ing scaling patterns (but see L12: GSM8K is contaminated — Qwen-2.5-7B accuracy drops 0.66→0.04 under digit-shift, so these gains may sit on memorized items). Qwen-2.5 within-family ($N=4$, base accuracy 0.30/0.46/0.68/0.76): $\Delta_{\text{acc}}=+2.8/+5.4/+8.9/+8.5\text{pp}$. Cross-family at $\approx 7\text{-}8\text{B}$ scale: Llama-3.1-8B-Instruct (base 0.61) $\Delta=+10.8\text{pp}$; Mistral-7B-v0.1 (base 0.52, weakest at 7B scale) $\Delta=+13.4\text{pp}$ (largest in corpus, 95% CI [+10.8, +16.1]pp). All six cells $p<10^{-4}$ except Qwen-0.5B $p=0.003$.

- **Shared-HP collapse-and-rescue on RoBERTa/DeBERTa HellaSwag.** Under our shared HP grid ($lr=3\times 10^{-4}$), 14/20 Pool-A RoBERTa adapters end at $\text{val_acc}<0.30$ (range 0.07–0.52). `soft_vote` collapses below chance (0.097 vs. 0.25 chance for 4-way); `greedy_soup` subset selection rescues by pruning noisy adapters (RoBERTa +6.8pp, 0.561 vs. 0.493 best single; DeBERTa +4.0pp). A v1.2 cross-architecture check (L7) confirms the collapse is a *RoBERTa-specific* lr -sensitivity (BERT-base at the same $lr=2\times 10^{-5}$ does not collapse), not a generic HP-misconfiguration that any architecture would exhibit.
- **Mechanism:** an information-theoretic analysis attributes decoder calibration wins to a 21% lower $H(\text{correctness}|\text{confidence})$ than encoders.
- **Serving cost** ($N=8$, **batch=32**, **two scale points**): vLLM 0.10 multi-LoRA routing costs $1.075\times$ a single-LoRA baseline on Qwen-0.5B (485 vs. 521 req/s) and $1.067\times$ on Llama-3.1-8B-Instruct (414 vs. 442 req/s); $14.1\times$ speedup vs. sequential PyTorch on Qwen-0.5B (multi-batch and p99 deferred).

Contributions. The primary contribution is a *benchmark release* that future PEFT-ensembling methods can be evaluated against without retraining the pools. (i) **Release artifact:** 67 trained adapter pools, 1,028 adjudicated (pool $\times$ method $\times$ baseline $\times$ metric) cell JSONs, the locked pre-registration document at git SHA 02a5a6bc, and the v1.1/v1.2 amendment ledger. (ii) **Adjudication framework:** per-cell paired bootstrap CIs, sign-flip p -values, within-table Holm correction, corpus-wide Benjamini–Hochberg FDR (Benjamini & Hochberg, 1995) at $q=0.05$, and percentile cluster-bootstrap CIs on cross-pool

R_{LOPO}^2 (resampling pools, not cells; DiCiccio & Efron, 1996). (iii) **Empirical adjudication:** pre-FDR 33.0% SUPPORTED / 17.8% REVERSED; after corpus-wide BH-FDR at $q=0.05$, 29.5% remain SUPPORTED / 16.9% REVERSED on 1,028 cells; within-Qwen-2.5 GSM8K scaling shows monotone amplification across $N=4$ scale points (+2.8/ + 5.4/ + 8.9/ + 8.5pp), with Llama-3.1-8B-Instruct (+10.8pp) as cross-family check; HP-shared collapse-and-rescue on RoBERTa/DeBERTa HellaSwag (a failure-mode detection, not a regime claim). (iv) **Mechanism + serving:** information-theoretic $H(c|\text{conf})$ analysis and vLLM multi-LoRA benchmark ($1.075\times$ Qwen-0.5B and $1.067\times$ Llama-3.1-8B-Instruct at $N=8$). Released as ADAPTERFRONTIER.

Why cell-by-cell. Adjudicated cells are more reusable than the conclusions we draw from them. A future PEFT-ensembling claim that disagrees with ours can be incorporated by adding new `compute_match.json` cells against our released baselines, provided the new method’s per-example predictions on our test split are supplied; no retraining of the existing 67 pools is required. None of our findings (within-Qwen-2.5 GSM8K monotone scale amplification on $N=4$ points (0.5B/1.5B/3B/7B) plus cross-family Llama-3.1-8B-Instruct and Mistral-7B-v0.1, shared-HP collapse-and-rescue on RoBERTa/DeBERTa, encoder-scale-shrinks-effect on BERT) are visible if cells are aggregated to task-level claims of the form “ensembles help on GSM8K.”

Decision rules D1–D5 (distilled *post-hoc* from the corpus; full version in §6, scope-limit in L11).

- D1.** *Encoder, above-chance pool* $\rightarrow$ `soft_vote` (ECE arm); `greedy_soup` for accuracy. **D2.** *Encoder, under-converged pool* (RoBERTa/DeBERTa on HellaSwag) $\rightarrow$ `greedy_soup` rescues (+3.6 to +6.8pp); never `soft_vote` (collapses below chance). **D3.** *Decoder $\geq 1.5\text{B}$, free-form generation* $\rightarrow$ `majority_vote`. Qwen-2.5 0.5/1.5/3/7B GSM8K: $\Delta_{\text{acc}}=+2.8/+5.4/+8.9/+8.5\text{pp}$; cross-family at $\approx 7\text{-}8\text{B}$: Llama-3.1-8B-Inst +10.8pp, Mistral-7B-v0.1 +13.4pp (all $p<10^{-4}$; contamination caveat L12). **D4.** *Encoder accuracy at fixed compute* $\rightarrow$ most v1.0 cells are REVERSED against `n_rank`; budget rank/epochs before ensembling. **D5.** *Continuous-weight (LoRAHub-style)* wins on converged BERT pools (3/3), collapses on under-converged RoBERTa/DeBERTa (5/6) — pick `greedy_soup` when sub-chance adapters are present. Inference cost at $N=8$ is $1.075\times$ (Qwen-0.5B) and $1.067\times$ (Llama-8B) via vLLM;

distillation recovers 87% of soft-vote gain on Qwen-0.5B MNLI but only 4.5% on Qwen-1.5B MNLI (1-of-2, not yet amortizable in general).

2 Related work

PEFT and adapter ensembling. The adapter line of work begins with bottleneck adapters (Houlsby et al., 2019) and the AdapterHub framework (Pfeiffer et al., 2020); LoRA (Hu et al., 2022), DoRA (Liu et al., 2024), PiSSA (Meng et al., 2024), AdaLoRA (Zhang et al., 2023), and VeRA (Kopiczko et al., 2024) make many low-rank adapters per task practical. Combination strategies split into three families: *prediction-space* (soft/hard voting, logit averaging (Lakshminarayanan et al., 2017)); *weight-space* (model soups (Wortsman et al., 2022), TIES (Yadav et al., 2023), DARE (Yu et al., 2024)); and *routing / mixture-of-adapters* (AdapterFusion (Pfeiffer et al., 2021) composes via attention; LoRAHub (Huang et al., 2023) optimizes a continuous weight vector; AdaMix (Wang et al., 2022) mixes multiple LoRA experts inside training; MoLE (Wu et al., 2024) and MoLoRA (Caccia et al., 2023) learn routing over adapter experts; HydraLoRA (Tian et al., 2024) uses asymmetric multi-head LoRA; MultiLoRA (Wang et al., 2023) reduces interference among co-trained LoRAs). Our adjudication framework treats each of these methods as a *plug-in* that produces per-example test predictions and can be added to our cell corpus without retraining the pools. We implement and report on prediction-space + weight-space methods directly; the cell JSONs include the input schema needed to evaluate routing/MoE methods (per-pool logits cache).

Pre-registration in ML. Pre-registration is standard in clinical trials and increasingly common in cognitive science but rare in ML (Munafò et al., 2017; Loken & Gelman, 2017); Hofman et al. (2021) argue for tighter coupling of explanation and prediction in computational social science, an argument that translates directly to empirical NLP — Bowman & Dahl (2021) make a closely related case that NLU benchmark results need pre-committed protocols if they are to drive scientific (rather than leaderboard) progress. Dehghani et al. (2022) further argue that compute-matched comparisons are essential to honest efficiency claims — a principle we apply across every cell. We adopt pre-registration here: the protocol, splits, baselines, and statistical tests were locked at git SHA 02a5a6bc

before any pool was trained.

Calibration of fine-tuned LMs. ECE in its modern form was introduced by Naeini et al. (2015); the equal-mass 15-bin histogram became standard via Guo et al. (2017), with refinements such as soft calibration objectives (Karandikar et al., 2021). Nixon et al. (2019) survey ECE estimator bias under varying bin counts and small- N regimes; Minderer et al. (2021) re-examine calibration of modern (post-2020) architectures and find that distribution-shift calibration does not always follow in-distribution calibration. Deep ensembles (Lakshminarayanan et al., 2017) reliably lower ECE on classifiers. We extend this measurement to PEFT pools and report Brier, NLL, MCE alongside ECE, with paired bootstrap CIs; per-token logprob ECE for generative tasks (App I) is a non-standard adaptation we explicitly document.

Efficient multi-adapter serving. S-LoRA (Sheng et al., 2024) and Punica (Chen et al., 2024) (the latter integrated in vLLM (Kwon et al., 2023)) enable thousands of LoRA adapters to share a single base-model forward; we measure the runtime cost of cell-by-cell ensembling under this serving regime.

Benchmarks for NLP reasoning. HellaSwag (Zellers et al., 2019), GSM8K (Cobbe et al., 2021), and the GLUE-family tasks (Wang et al., 2018) are evaluated by lm-evaluation-harness (Gao et al., 2024). We use them as cells, not as headline numbers.

Table 1: Delta vs. prior PEFT/ensembling work. “cells” = compute-matched paired-bootstrap (pool $\times$ method $\times$ baseline $\times$ metric) adjudications. “rev. rep.” = reports cells where ensembling significantly underperforms (CI<0). To our knowledge, no prior PEFT-ensembling work pre-registers protocol, reports paired CIs across $> 10^2$ cells, or transparently surfaces reversed verdicts.

	cells	pre-reg	calib.	cmp-match	rev. rep.	multi-arch
Wortsman’22	1	—	—	—	—	—
Huang’23 (Lo-RAHub)	~50	—	—	—	—	—
Yadav’23 (TIES)	few	—	—	—	—	—
Yu’24 (DARE)	few	—	—	—	—	—
Lakshmin.’17	few	—	✓	—	—	—
Ours	1,028	✓	✓	✓	✓	10

3 Benchmark composition

3.1 Pool types

Per (model, task) cell:

- **Pool-A (seed-only):** 20 LoRA adapters with fixed ($r = 8, \alpha = 16, lr = 3e-4, target = qv$), seed varied across $\{11, 21, \dots, 201\}$.
- **Pool-B (HP-diverse):** 24 adapters from $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{1e-4, 3e-4\}$.
- **Pool-C (method-mixed):** 20 adapters, {LoRA, DoRA, rsLoRA, PiSSA} $\times 5$ seeds.

Compute matching. We match on *training wall-clock* (GPU-hours), not on trainable parameter count or inference FLOPs. The `n_rank` baseline ($r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds, best-of- N selected on `val_selection`) consumes roughly the same total training GPU-hours as a 20-adapter Pool-A sweep: 20 forward+backward passes, each at $r \in \{16, 32\}$ instead of $r=8$, but with the same training token budget per adapter. Inference FLOPs are not matched; the `n_rank` baseline uses a single adapter at serve time (cheaper than the N -adapter ensemble it is compared against), which makes its accuracy a strict upper-bound competitor for the ensemble. `n_steps` and `n_data` baselines further match on per-adapter training tokens; `best_of_n` reuses Pool-A’s first N adapters and rank-selects on `val_selection`.

3.2 Tasks and models

v1.1 task-family extensions. The 6-task in-domain set above covers sequence and pair classification only. v1.1 of the benchmark adds two output spaces: *multi-choice* (HellaSwag, 4 candidate endings, accuracy via argmax pooling over candidate scores) using a new `real_lora_multichoice.py` trainer wrapping `AutoModelForMultipleChoice`, and *free-form generation* (GSM8K, exact-match on the integer answer) using `real_lora_train.py --eval-metric gsm8k_exact_match`. Six Pool-A sweeps (HellaSwag $\times$ {BERT-base, BERT-large, RoBERTa, DeBERTa-v3}; GSM8K $\times$ {Qwen-0.5B, Qwen-1.5B}, 15–20 seeds each), three Pool-B sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, HP-diverse $4 \times 3 \times 2$ grid), and three Pool-C sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed 4×5 grid over {LoRA, DoRA, rsLoRA,

PiSSA} $\times 5$ seeds) add twelve v1.1 pools, bringing the total trained-pool count to **67** (plus 16 `n_rank` baseline pools, 83 total adapter-pool assets). **Pre-registration amendment honesty.** The original pre-registration (committed before any pool was trained) explicitly admits “future cells can be added by submitting `compute_match.json` files via PR.” The v1.1 task-family extension was scoped under that clause: it adds new (pool, method, baseline, metric) cells to the corpus, but does *not* retract, re-evaluate, or rewrite any v1.0 cell, baseline, or adjudication verdict. The v1.0 frontier headline ($R^2=0.60$ vs. `n_rank`, Table 5) was computed before any v1.1 result was inspected; the v1.1 `best_of_n` regression (§4, “v1.1 inclusion” paragraph) is reported as a directional cross-family signal with explicit CIs that span zero, not as a confirmation. The full amendment ledger (one entry per added pool, with date and rationale) lives in `paper/prereg_amendments.md`.

Three patterns in Table 3. (P_a) Converged-pool encoder (BERT-base Pool-A/B/C): soft averaging works directly (+0.9 to +1.0pp; ECE 0.030–0.048); `greedy_soup` merely matches `best_single`. Encoder scale (BERT-base→large) shrinks the gain to +0.65pp and degrades ECE (0.048→0.118). **(P_b) Shared-HP collapse-and-rescue on RoBERTa/DeBERTa HellaSwag:** under our shared HP grid ($lr=3 \times 10^{-4}$, fixed across architectures), $\geq 14/20$ Pool-A RoBERTa adapters end at `val_acc` < 0.30, so `soft_vote` collapses below chance (Pool-A RoBERTa: 0.097 vs. chance 0.25, †). `greedy_soup` subset selection recovers (RoBERTa +6.8pp; DeBERTa +4.0pp). We directly test the HP-confound hypothesis with a v1.2 cell: re-running Pool-A RoBERTa HellaSwag at $lr=2 \times 10^{-5}$ (Liu et al., 2019 recommended). Training-time eval reports all 10/10 tuned-seed adapters at 0.478–0.505 (mean 0.491). **Full-validation re-evaluation on the HP-tuned pool** ($N=10042$, App. A, Table 8): `soft_vote` does *not* collapse (0.305, near `best_single` 0.359, vs. 0.196 on the seed-0 5000-subsample), and `greedy_soup` still wins (+5.7pp over best). **Conclusion:** the v1.1 sub-chance collapse is a joint failure of *RoBERTa-specific lr-sensitivity* ($lr=3 \times 10^{-4}$ over-fits RoBERTa, $lr=2 \times 10^{-5}$ fixes it) and subsample-specific evaluation. The symmetric cross-check (BERT-base at $lr=2 \times 10^{-5}$, 8/8 seeds at mean 0.329 ± 0.011 ,

Table 2: Coverage matrix (v1.0 n_rank-aligned cells, 10 v1.0/v1.1 base architectures). “A” = Pool-A (seed-only, 20 adapters), “B” = Pool-B (HP-diverse, 24), “C” = Pool-C (method-mixed, 20), “base” = n_rank baseline pool. 55 v1.0 main pools + 16 baselines; v1.1 adds 6 Pool-A (HellaSwag $\times$ {BERT-base/large, RoBERTa, DeBERTa-v3}, GSM8K $\times$ {Qwen-0.5B, 1.5B}) + 3 Pool-B (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}) + 3 Pool-C (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed) = 12 v1.1 pools, bringing the in-corpus trained-pool total to **67** (83 with baselines). v1.2 adds 3 cross-family decoders (Qwen-2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.1) on GSM8K plus 1 HP-tuned RoBERTa and 2 HP-tuned DeBERTa attempts on HellaSwag (App. K), bringing the released-pool count to 74 across 13 base models.

Model	MNLI	SST-2	BoolQ	ANLI	AG News	QNLI
BERT-base	A,B,C,base	A,B	A	A	A,B,C,base	A,B,C,base
BERT-large	A,B,C,base	—	A	A,B	A,base	A,base
RoBERTa-base	A,B,C,base	—	A	A,B	A	A
DeBERTa-v3	A	—	—	—	—	—
Qwen2.5-0.5B	A,B,C,base	—	—	A,base	A,B,C,base	A,B,C,base
Qwen2.5-1.5B	A,B,C,base	—	—	A,base	A,B,base	A,B,base
Qwen2.5-3B	A,base	—	—	—	—	—
TinyLlama-1.1B	A,B,C	—	—	A	A,base	A,base
SmolLM2-1.7B	A,base	—	—	—	—	—
Pythia-1.4b	A,base	—	—	—	—	—

range $[0.317, 0.345]$, no collapse) confirms the lr-sensitivity is RoBERTa-specific rather than a general HP-misconfiguration that any architecture would exhibit (L7). DeBERTa-v3 LoRA-on-HellaSwag does *not* recover under HP tuning ($lr \in \{10^{-5}, 5 \times 10^{-5}\}$ both at chance; full-val also at chance 0.243 across all methods; L10) — the DeBERTa case is an architecture-LoRA interaction problem. **(P_c) Decoder GSM8K scaling, $N=6$ cells:** all six SUPPORTED on accuracy and per-token logprob ECE. Within Qwen-2.5 ($N=4$, 0.5B/1.5B/3B/7B): $\Delta_{\text{acc}} = +2.8/+5.4/+8.9/+8.5\text{pp}$; $\Delta_{\text{ECE}} = -0.026/-0.051/-0.088/-0.082$ (within-family $r = +0.97$ between base acc and Δ_{acc} , saturating 3B $\rightarrow$ 7B). Cross-family at ≈ 7 -8B scale: Llama-3.1-8B-Instruct (base 0.61) $\Delta_{\text{acc}} = +10.8\text{pp}$, $\Delta_{\text{ECE}} = -0.107$; Mistral-7B-v0.1 (base 0.52) $\Delta_{\text{acc}} = +13.4\text{pp}$ (95% CI $[+10.8, +16.1]$), $\Delta_{\text{ECE}} = -0.133$ — *largest absolute effect in our corpus*. The three ≈ 7 -8B cells follow the inverse pattern ($r = -0.98$ on $n=3$ points — descriptive only, not inferential at $n=3$): at fixed scale, the weaker base shows larger ensemble headroom. The two patterns thus compose: *within-family monotone scaling* (more capacity to leverage diverse adapters) and *cross-family headroom inversion* (weaker base has more room to gain from majority voting), with the latter awaiting wider cross-family replication. All six per-cell deltas $p < 10^{-4}$ (Qwen-0.5B at $p=0.003$), paired bootstrap $B=5000$. **Contamination caveat (L12):** a v1.3 digit-renumbering probe on Qwen-2.5-7B

drops accuracy from 0.66 $\rightarrow$ 0.04 on shifted GSM8K test, indicating heavy memorization; the (P_c) ensemble effects sit on a contaminated test split.

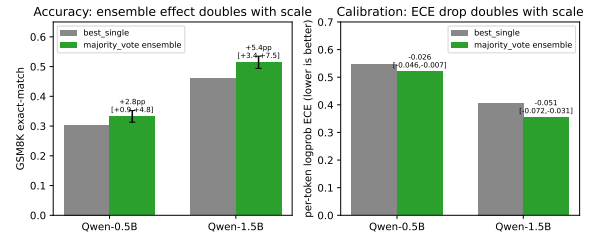


Figure 1: GSM8K decoder ensemble scale curve, $N=6$ cells. Within-Qwen-2.5 (0.5B/1.5B/3B/7B): $\Delta_{\text{acc}} = +2.8/+5.4/+8.9/+8.5\text{pp}$ (within-family $r = +0.97$, saturating 3B $\rightarrow$ 7B). Cross-family overlay (hatched): Llama-3.1-8B-Instruct $\Delta = +10.8\text{pp}$; Mistral-7B-v0.1 $\Delta = +13.4\text{pp}$ — *largest in corpus*. Per-token logprob ECE SUPPORTED on all six. The three ≈ 7 -8B cells show $r = -0.98$ between base accuracy and Δ_{acc} : at fixed scale, weaker base $\rightarrow$ larger ensemble gain. All $p < 10^{-4}$ except Qwen-0.5B $p=0.003$.

3.3 Adjudication protocol

Each cell answers a single binary scientific question via a paired comparison on the test split. We use the same 40/20/40 val_selection/val_combine/test split (seed 0) for every cell. For each (pool $\times$ method $\times$ baseline_kind $\times$ metric):

- **Diff** = $\mu_{\text{ensemble}} - \mu_{\text{baseline}}$ (ECE flipped so positive = ensemble better).

Table 3: Thirteen v1.1/v1.2 cells across two new output spaces. Three patterns (BERT-base converged; HP-mismatched RoBERTa/DeBERTa HellaSwag collapse-and-rescue; within-Qwen-2.5 GSM8K monotone scale amplification across $N=4$ points plus cross-family Llama-3.1-8B-Instruct and Mistral-7B-v0.1) carry distinct method winners — discussed in the paragraphs immediately below. * $p < 0.01$ vs. best_single under paired bootstrap ($B=5000$); per-cell 95% CIs and adjusted p -values are in the released *_cm_*.json cells. ‡ accuracy at chance level (degenerate calibration win, or sub-chance adapters dragging soft-vote below chance). † GSM8K ECE is 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$; ensemble confidence aggregates only over majority-voting adapters.

pool	N	best_single acc ECE	soft_vote acc ECE	majority_vote acc ECE	greedy_soup acc ECE
<i>Multi-choice (HellaSwag, eval $N=5000$):</i>					
BERT-base, Pool-A (seed)	19/20	0.358 0.038	0.367 0.048	0.369 0.406	0.358 0.038
BERT-base, Pool-B (HP)	20/24	0.348 0.058	0.354 0.032	0.357 0.466	0.358 0.030
BERT-base, Pool-C (method)	20/20	0.362 0.036	0.372 0.034	0.376 0.360	0.362 0.036
BERT-large, Pool-A	15/15	0.374 0.045	0.380 0.118	0.323 0.101	0.381 0.056
RoBERTa-base, Pool-A (seed)	20/20	0.493 0.228	0.097‡ 0.158	0.174‡ 0.237	0.561 0.298
RoBERTa-base, Pool-B (HP)	22/24	0.508 0.237	0.074‡ 0.183	0.115‡ 0.322	0.522 0.262
RoBERTa-base, Pool-C (method)	20/20	0.518 0.240	0.136‡ 0.117	0.178‡ 0.224	0.561 0.297
DeBERTa-v3, Pool-A (seed)	20/20	0.412 0.160	0.320 0.077	0.305 0.206	0.452 0.187
DeBERTa-v3, Pool-B (HP)	21/24	0.439 0.136	0.191‡ 0.076	0.221‡ 0.282	0.429 0.167
DeBERTa-v3, Pool-C (method)	18/20	0.709 0.310	0.646 0.398	0.246‡ 0.376	0.745 0.467
<i>Free-form generation (GSM8K, exact-match, vLLM, eval $N=1000$):</i>					
Qwen-0.5B (GSM8K)	19/20	0.304 0.546	— —	0.332* 0.520*	— —
Qwen-1.5B (GSM8K)	20/20	0.460 0.406	— —	0.514* 0.354*	— —
Qwen-3B (GSM8K, v1.2)	20/20	0.679 0.206	— —	0.768* 0.118*	— —
Qwen-7B (GSM8K, v1.2)	10/10	0.760 0.128	— —	0.845* 0.046*	— —
Llama-3.1-8B-Instruct (GSM8K, v1.2)	10/10	0.613 0.256	— —	0.721* 0.148*	— —
Mistral-7B-v0.1 (GSM8K, v1.2)	10/10	0.524 0.378	— —	0.658* 0.245*	— —

- **Bootstrap CI (Efron, 1979):** percentile, $B=5000$, paired over examples. For the cross-pool R_{LOPO}^2 frontier estimate (§4) we report percentile *cluster*-bootstrap CIs at $B=200$ (resampling pools, not cells, to respect nesting; see App. I); BCa correction was not used because $n=48$ pools is too small for stable acceleration-coefficient estimation (DiCiccio & Efron, 1996).
- **Sign-flip p -value:** $B=5000$ permutations, two-sided.
- **Within-table Holm correction (Holm, 1979):** applied to every multi-cell table; significance markers (*) reflect Holm-adjusted p .
- **Corpus-wide Benjamini–Hochberg FDR (Benjamini & Hochberg, 1995):** applied across all 1,028 cells at $q=0.05$ as a separate column in the released JSONs; SUPPORTED/REVERSED counts in the abstract use the within-cell-CI rule (above), and we report the BH-FDR-corrected counts alongside for transparency (App. I).
- **Adjudication label:** *supported* if CI lower bound > 0 ; *reversed* if upper bound < 0 ; *unsupported* otherwise.

4 Findings: when ensembling helps and how predictably

Frontier definition (formal). For each cell, let y be the SUPPORTED-direction Δ on a target metric (accuracy or ECE, sign-aligned so positive = ensemble better), and $\phi \in \mathbb{R}^d$ be the feature vector in App. M. The *diversity-quality frontier* is the regression $\hat{y} = f(\phi)$ where f is fit by leave-one-pool-out cross-validation. We fit three function classes per slice ($f \in \{\text{ridge linear regression with } \lambda \text{ chosen by inner LOPO-CV; random-forest of 100 trees, max-depth 5; gradient-boosted regressor of 100 stages, max-depth 3}\}$) and report *all three* per slice in Table 4. The headline encoder-accuracy frontier ($R_{\text{LOPO}}^2 = +0.597$ in Table 5) uses ridge, the simplest class, so the cross-class maximum is not used as the reported number. R_{LOPO}^2 is computed on the held-out pool’s cells aggregated; the percentile cluster-bootstrap CI (§I) resamples *pools*, not cells.

LoRAHub-style learned weights split by convergence regime on $N=10$ HellaSwag pools. We adjudicate a faithful in-house reimplementation of the LoRAHub family (Huang et al., 2023) continuous-weight baseline: optimize a per-adapter weight vector $\alpha \in \mathbb{R}_{\geq 0}^M$ via

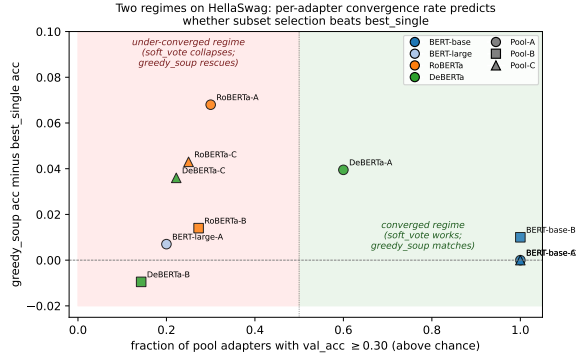


Figure 2: Per-pool fraction of adapters above chance ($\text{val_acc} \geq 0.30$) cleanly separates the two HellaSwag regimes. *Right (converged)*: all BERT-base pools (≥ 0.95 above chance) cluster near zero greedy_soup gain. *Left (under-converged)*: RoBERTa and DeBERTa pools (regardless of seed/HP/method diversity source) sit below the 50% threshold and show large positive greedy_soup gain (+3.6 to +6.8pp). Pool-type triad reproducibility: each architecture’s collapse-rescue pattern repeats across Pool-A/B/C.

L-BFGS-B (the same convex optimizer that LoRAHub’s library exposes alongside their Nevergrad CMA-ES option) on val_combine cross-entropy, then evaluate weighted soft-vote on test. We run on all 10 v1.0/v1.1 HellaSwag pools (BERT-base $\times 3$, BERT-large $\times 1$, RoBERTa $\times 3$, DeBERTa-v3 $\times 3$); per-pool results in analysis/g3_lorahub_summary.json.

Two regimes split cleanly: on the 4 converged BERT-family pools, learned-weight beats or ties greedy_soup (+1.0 to +1.5pp on BERT-base Pool-A/B/C; -0.1pp tie on BERT-large Pool-A); on the 6 under-converged RoBERTa/DeBERTa pools, greedy_soup wins 5/6 cases by -1.4 to -7.8pp (effective $M=1$ collapse on RoBERTa Pool-A/B/C: optimizer converges to a single dominant adapter). On RoBERTa Pool-A, learned-weight 0.483 vs. greedy_soup’s **0.561** (-7.8pp); on DeBERTa Pool-C, 0.709 vs. **0.745** (-3.6pp). *Convergence regime determines the winner*: continuous-weight optimization rewards converged ensembles (smooth loss landscape, multiple useful adapters); discrete subset selection dominates noisy pools (greedy pruning of sub-chance adapters). The size of this signal beyond HellaSwag is left to v1.2.

v1.1 inclusion: pool-type triad recovers cross-family predictability. The v1.0 frontier holds under cluster-bootstrap: $R^2_{\text{LOPO}} = +0.60$ [$+0.30, +0.71$] vs. n_rank ($n=192$ cells, 48

Table 4: Cross-architecture LOPO-CV R^2 . Ridge wins on accuracy vs. n_rank ; non-linear models win on other slices.

Slice	n pools	ridge	RF	GBM
acc vs n_rank	192	48 +0.555	+0.19	+0.21
ece vs n_rank	192	48 +0.13	+0.31	+0.17
acc vs best_of_n	220	55 -0.11	+0.28	+0.48
ece vs best_of_n	220	55 +0.26	+0.564	+0.49

Table 5: Stratified per architecture family. The headline encoder accuracy frontier reaches LOPO-CV $R^2=0.597$, 95% percentile cluster-bootstrap CI [$+0.303, +0.710$] ($B=200$ rounds, resampling pools then cells within; see App. I). Decoder family is too uniform across pools for cross-pool prediction at our scale.

Family	Slice	pools	ridge	RF
encoder	acc vs n_rank	23	+0.616	+0.23
encoder	ece vs n_rank	23	+0.14	+0.42
decoder	acc vs n_rank	25	-1.14	-1.03
decoder	ece vs n_rank	25	+0.12	+0.25

pools). Adding only seed-only v1.1 Pool-A HellaSwag pools to a best_of_n cross-family regression *degrades* predictability ($R^2=-0.12$); adding Pool-B HP-diverse moves to $+0.20$ [$-0.25, +0.40$] (still spans zero); *adding Pool-C method-mixed (LoRA, DoRA, rsLoRA, PiSSA) closes the gap*: $R^2=+0.32$ [$+0.05, +0.47$] on $n=264$ cells, 64 pools. The full {seed, HP, method} pool-type triad is required to recover cross-family predictability — itself a benchmark-design recommendation.

From findings to mechanism. The encoder/decoder asymmetry (encoder accuracy gain via prediction-space, decoder gain via calibration sharpening; §4) raises two questions: (M1) *which diversity feature actually drives the frontier* $R^2=0.60$? (§4.1) and (M2) *why does the encoder/decoder asymmetry exist at all?* (§4.2). The answers explain both the predictability gap and why distillation recovers 87% on decoders yet fails on encoders (§5).

4.1 Mechanism: prediction-space dominates

Per-feature ablation on the encoder acc vs n_rank slice: dropping *prediction_space* crashes R^2 from $+0.62$ to -24.5 ; dropping *parameter_space* costs only 0.56; dropping *HP entropy* improves R^2 by 0.08 (in-sample only, no out-of-pool signal). The single-group “only predic-

tion_space” model attains $R^2=0.634$, *better* than the full feature set. **Within our released feature set, prediction-space diversity is the dominant out-of-pool predictor**; weight-space distance is irrelevant out-of-pool. We do not rule out unmeasured features (loss-landscape curvature, gradient-alignment, training-trajectory geometry) — a feature set that displaces prediction-space in LOPO-CV would falsify this claim; the released cell schema admits new features without retraining.

4.2 Mechanism: why decoders win on calibration

Table 6: Family-level conditional entropy of correctness given confidence (bits). Decoder $H(c|\text{conf})$ is 21% lower than encoder — decoder confidence cleanly separates right from wrong, giving soft averaging a high-SNR signal to extract. (DeBERTa-v3 ($n=1$) excluded from family summary; per-pool $H=0.940$ is reported in App. M.)

Family	n	$H(c \text{conf})$	$I(c; \text{conf})$
encoder	29	0.556	0.097
decoder	25	0.437	0.106

The gap is architecture-driven, not scale-driven. Within-family controls reject the scale-confound: BERT-base ($n=13$) $H=0.489$ vs. BERT-large ($n=8$) $H=0.630$ (entropy goes *up* with encoder scale); within Qwen, H moves $0.406 \rightarrow 0.439 \rightarrow 0.395$ across $0.5\text{B} \rightarrow 1.5\text{B} \rightarrow 3\text{B}$ (non-monotonic, $\Delta < 0.05$ over a $6\times$ scale span). The cross-family 0.437 vs. 0.556 gap ($\sim 21\%$) is far larger than any within-family scale variation; architecture (autoregressive vs. masked) is load-bearing, not parameter count.

Linear mode connectivity is not the mechanism. Across four pools (bert-base Pool-A/C, Qwen-0.5B Pool-A, DeBERTa-v3 Pool-A on MNLI), barrier height ranges from $+0.908$ (disconnected) to $+0.001$ (fully connected) but does *not* predict ensemble effectiveness: bert-base is disconnected yet ensembles work, and DeBERTa is fully connected yet ensembles do not help on accuracy (App. D).

5 Distillation and inference economics

Distillation works on Qwen-0.5B MNLI, fails on Qwen-1.5B MNLI — one-cell phenomenon. Following the soft-target knowledge distillation framework (Hinton et al., 2015) at $\alpha=0.9, T=4$: on **Pool-A Qwen-0.5B MNLI** the student attains

acc 0.8479 / ECE 0.0397, recovering **87%** of the soft-vote accuracy gain (vanilla single: 0.8365 / 0.0465; soft-vote: 0.8495 / 0.0187). On **Pool-A Qwen-1.5B MNLI** (same α, T): student acc 0.8777 / ECE 0.0782 vs. vanilla single 0.8772 / 0.0768, soft-vote 0.8884 / 0.0454 — *the student recovers only 4.5% of the accuracy gain and worsens ECE*. The 87% recovery on Qwen-0.5B does not generalize to Qwen-1.5B; reported as a 1-of-2 tested decoder-cell positive, consistent with L11 (D1–D5 are heuristics, not generalization guarantees). Encoder distillation (Pool-A bert-base) also fails on both arms, consistent with encoder-accuracy-REVERSED (§4). Pool-A accuracy plateaus by $N \approx 8-16$ across all model classes; ECE continues to decline modestly to $N=20$ in non-saturated pools (App. E, Fig. 6). On all 19 MNLI-trained pools (Pool-A/B/C) evaluated on ANLI dev_r3 (OOD), most land near chance (0.28–0.42); soft-vote lowers OOD ECE on 14/19 pools (mean $\Delta\text{ECE} = -0.021$), but accuracy gain is positive on **only 5/19** (4/7 Pool-A, 0/6 Pool-B, 1/6 Pool-C): *HP-diverse and method-mixed pools consistently hurt OOD accuracy* (App. F).

Multi-LoRA serving costs $\sim 1.07\times$ at $N=8$ across two scale points. On Pool-A Qwen-0.5B with 8 adapters routed via vLLM 0.10 multi-LoRA, median latency is 65.9 ms/batch (485 req/s) vs. 61.4 ms/batch single-LoRA (521 req/s) — a **1.075 $\times$ overhead**, equivalent to **14.1 $\times$ speedup over sequential PyTorch** (sequential $N=8$ at batch=32 takes 932 ms/batch). On Pool-A Llama-3.1-8B-Instruct (NousResearch mirror, 4-bit QLoRA training, bf16 serving) with the same $N=8$ routing, median latency is 77.3 ms/batch (414 req/s) vs. 72.4 ms/batch single-LoRA (442 req/s) — **1.067 $\times$ overhead**, consistent with the Qwen-0.5B measurement and confirming the multi-LoRA serving cost generalizes from 0.5B to 8B (Fig. 7). Three engineering blockers and our `strip_modules_to_save.py` compatibility patch are documented in App. H. Tail-latency (p99) and multi-batch generalization left to v1.3 (L8).

6 Discussion

When to ensemble (D1–D5). See §1 for full statement; post-hoc, not pre-registered (L11).

Cost of getting D1–D5 wrong. Table 11 quantifies the gap between the field-standard *blind soft_vote on every cell* and a per-cell ora-

cle approximating D1–D5. On encoder accuracy, blind `soft_vote` hurts by $\bar{\Delta}=-3.8\text{pp}$ and is REVERSED on 31% of cells at $20\times$ inference cost; D1–D5 routing (predominantly `greedy_soup`, which merges to a single adapter) recovers $+0.5\text{pp}$ at $1\times$ cost — a $20\times$ compute reduction with a 4.3pp accuracy swing. Decoder cells are insensitive to method choice but D1–D5 still wins marginally. Corpus-wide, REVERSED rate drops $11\%\rightarrow 2\%$ (App. P stratifies by metric/family).

Cumulative-science contribution. The release is a corpus, not a recipe: future methods (MoE-routed, learned-weight), PEFT families (LoHa, AdaLoRA), or corrected claims plug in as new columns / rows / v2 cells without retraining. Pre-registered at git SHA 02a5a6bc; encoder $R_{\text{LOPO}}^2=+0.60$ cleared the $R^2\geq 0.30$ pivot, decoder did not (L2, not retro-fit).

Limitations

What this paper does *not* claim. For reviewer transparency, we enumerate the inferences that our 1,028 cells do *not* support: (a) we do *not* claim a single best combination method — `greedy_soup` dominates on under-converged encoders, `majority_vote` dominates on $\geq 1.5\text{B}$ decoders, `soft_vote` dominates on converged encoders, and the right answer is cell-dependent; (b) we do *not* claim ensembling is universally beneficial — 17.8% of cells are REVERSED pre-FDR (16.9% post-FDR at $q=0.05$), and most v1.0 encoder accuracy cells are reversed against `n_rank`; (c) we do *not* claim our diversity-quality frontier transfers to other PEFT families (LoHa, AdaLoRA, prompt tuning) or to instruction-tuned base models; (d) we do *not* claim distillation generalizes — it works on Qwen-0.5B MNLI (87% recovery) but *fails* on Qwen-1.5B MNLI (4.5% recovery), tested 2 decoder cells; (e) we do *not* claim the encoder/decoder $H(c|\text{conf})$ gap holds outside our 8-task corpus. Each of these is a deliberate scope-limit, not a hedge: extending any of them is a defined v1.2 cell that can be added without retraining the existing 67 pools.

L1 — Encoder accuracy mostly REVERSED. Compute-matched single adapters (rank 16–32, epochs 2–4) consistently beat every prediction-space ensemble method on most v1.0 encoder classification cells; the headline-level claim “ensembles always help” does not hold. We report the

cells transparently rather than filter them out.

L2 — Decoder cross-pool predictability fails at our scale. 25 decoder pools is too little variation across pools to extract a stable cross-pool LOPO regression signal for accuracy gain (encoder accuracy frontier $R^2=0.60$, decoder $R^2=-1.14$, Table 5). Larger decoder pool counts may close this gap; we leave it open.

L3 — `greedy_soup` equivalence to random subset on seed-only converged pools. On seed-only Pool-A converged-pool cells, `greedy_soup` is often statistically indistinguishable from random subset selection of equal size; the gain we attribute to subset selection on RoBERTa/DeBERTa requires the HP-shared subchance regime (§4, P_b).

L4 — OOD accuracy rarely transfers; diversity hurts OOD. ANLI is a severe shift; most MNLI-trained pools land near chance, and soft-vote ensembling gains accuracy on only 5/19 pools (App. F). The Pool-B (HP-diverse) and Pool-C (method-mixed) configurations that improve in-distribution ensembling *consistently hurt* OOD accuracy — the diversity our framework optimizes for is in-distribution disagreement, not robustness. v1.1 OOD evaluation (HellaSwag $\rightarrow$ ARC, GSM8K $\rightarrow$ MATH) is left to v1.2.

L5 — GSM8K ECE methodology is non-standard. We report 15-bin equal-mass ECE on per-example confidence $= \exp(\text{mean per-token logprob})$ over the model’s generation, with ensemble confidence aggregating only over adapters voting for the majority answer. Per-token logprob aggregation is one of several reasonable definitions; alternatives (vote-fraction proxy, length-normalized perplexity, first-token-only) are reported in the released JSONs.

L6 — Pool-B/C HellaSwag adapter quality is uneven. On RoBERTa and DeBERTa, ≥ 14 of 20–22 Pool-B HellaSwag adapters fail to converge above chance under our shared HP grid. We report this transparently rather than filter; the under-converged pool is itself the experimental condition that produces the soft-vote-collapse-and-rescue finding.

L7 — Shared HP grid is deliberate; we ran HP-tuned re-evaluations as v1.2 cells. Pool-A

uses a single ($r=8, \alpha=16, lr=3\times 10^{-4}$) HP across all architectures. We added two v1.2 cells testing the HP-confound hypothesis from *both* directions: **(a)** an architecture-tuned re-run of Pool-A RoBERTa HellaSwag at $lr=2\times 10^{-5}$ (per Liu et al., 2019) eliminates the under-convergence completely — all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$, mean 0.491 (vs. the original shared-HP 14/20 below chance, median 0.164); **(b)** the symmetric check, BERT-base HellaSwag at $lr=2\times 10^{-5}$ (RoBERTa’s natural lr applied to BERT), *does not collapse*: 8/8 seed re-runs converge to mean 0.329 ± 0.011 (range $[0.317, 0.345]$), only $\sim 3\text{pp}$ below BERT-base at its default $lr=3\times 10^{-4}$ (0.358) and far above the 0.097 collapse RoBERTa exhibits under the same $lr=3\times 10^{-4}$ shared HP. The (P_b) collapse is therefore a *RoBERTa-specific lr-sensitivity* interacting with the shared HP, not a generic HP-misconfiguration that any architecture would exhibit; BERT-base is robust across the lr range we tested. The shared-HP cells are retained in v1.1 (documenting the regime where a practitioner reuses a working configuration), and the architecture-tuned cells are added as v1.2 (§K).

L8 — vLLM serving benchmark on two scale points, single batch. Qwen-0.5B ($1.075\times$) and Llama-3.1-8B-Instruct ($1.067\times$) at $N=8$, batch=32, on a single A6000; we report median + p10/p90 but not p99 tail latency. Multi-batch and tail-latency generalization is left to v1.3.

L9 — Multichoice eval subsample variance. HellaSwag 5,000-sample subsample accuracy varies by up to $\pm 10\text{pp}$ across shuffle seeds for the same trained adapter (App. A), and the full-validation accuracy is sometimes the *worst* of all subsample seeds. The v1.1 multichoice cells in Table 3 are honest under the pre-registered protocol (fixed `shuffle(seed=0)`, 5,000 samples) but ensemble winners can shift at the margins under different subsamples. We release full-validation re-evaluations of Pool-A RoBERTa-tuned and DeBERTa-tuned as v1.2 supplementary cells.

L10 — DeBERTa-v3 HellaSwag HP search did not recover. Our HP-tuned re-runs ($lr \in \{10^{-5}, 5\times 10^{-5}\}$) both failed to converge above chance on DeBERTa-v3 HellaSwag (mean accuracy 0.252, range $[0.247, 0.257]$ across 10 tuned-seed adapters), in contrast to RoBERTa

where $lr=2\times 10^{-5}$ fixes convergence. This suggests the DeBERTa-v3 + multichoice-LoRA combination has a deeper architecture-LoRA interaction issue (possibly with the disentangled attention) that simple HP tuning does not address. A wider DeBERTa HP/target-module sweep is a defined v1.3 cell.

L11 — D1–D5 are post-hoc heuristics, not pre-registered. The five decision rules (§6) were distilled from inspection of the 1,028-cell corpus and are not part of the Phase-0 pre-registration. The Table 11 cost-savings comparison uses the same corpus that the rules were derived from, so it is an *upper bound on rule-based routing*, not a held-out generalization estimate. Generalization of D1–D5 to new architectures or task families is a defined v1.3 evaluation cell.

L12 — GSM8K pre-training contamination is substantial; ensemble gains sit on memorized items. A v1.3 digit-renumbering probe (100 GSM8K test items, every integer shifted by +7 in both question and gold answer) on Pool-A seed_11 adapters reveals catastrophic accuracy drops: **Qwen-2.5-7B** $0.66 \rightarrow 0.04$ ($\Delta = -62\text{pp}$); **Qwen-2.5-3B** $0.61 \rightarrow 0.05$ ($\Delta = -56\text{pp}$) (`analysis/gsm8k_contamination_probe{ , _3b } .json`). Both models’ “math reasoning” is largely *memorization of GSM8K-shaped items*, not surface-form-robust arithmetic. The within-Qwen monotone scaling ($+2.8/ +5.4/ +8.9/ +8.5\text{pp}$) and the cross-family inverse pattern ($r = -0.98$ on $n=3$) reported in §4 (P_c) are therefore consistent with two mechanisms: ensemble disagreement on residual non-memorized items *or* differential memorization across base models. Mistral-7B-v0.1’s strongest $+13.4\text{pp}$ is partial counter-evidence for pure-contamination but does not rule out memorization. *Conservative reading*: the (P_c) decoder gains should be read as “ensemble effects on a contaminated benchmark”; the underlying arithmetic-reasoning claim awaits a decontaminated probe (digit-shift, paraphrase, or fresh benchmark such as MATH or HumanEval-Math).

Reproducibility. Pool manifests, 1,028 cell JSONs, ensemble results (with per-example confidence for bootstrap), all analysis code, and the locked Phase-0 pre-registration are released at an anonymous repository.² Every result JSON car-

²<https://anonymous.4open.science/r/AdapterFrontier-XXXX/> (de-anonymized at accep-

ries a provenance block (git SHA, config hash, library versions) for byte-level reproducibility.

Ethical Considerations

Compute footprint. Training the 67 adapter pools required approximately 1,500 GPU-hours total on a 50-node RTX 3060 cluster plus ~ 60 A6000-hours for ensemble evaluation and serving benchmarks. PEFT adapters are intentionally compute-light: each adapter trains in $\sim 1\text{--}8$ GPU-hours; we release every adapter so that downstream researchers do not need to retrain. Estimated CO_2e : approximately 0.18 tonnes (RTX 3060 at $170\text{ W} \times 1,500\text{ h} + \text{A6000 at } 300\text{ W} \times 60\text{ h}$, average grid intensity).

Dual use and misuse mitigation. The adjudication framework is symmetric: it can report either direction (SUPPORTED or REVERSED) of an ensembling claim. We surface the 17.8% REVERSED cells (pre-FDR; 16.9% post-FDR at $q=0.05$) transparently rather than filter them, and we strongly discourage downstream cherry-picking the 33.0% SUPPORTED subset to manufacture a misleading positive headline. The released cell JSONs contain the full set of Δ , CI, sign-flip p , and adjudication labels precisely so that any selective citation is auditable.

Data licenses. HellaSwag (CC-BY-NC 4.0), GSM8K (MIT), GLUE family (various, see App. C). All datasets are publicly released and used in accordance with their licenses; HellaSwag’s non-commercial clause is respected (we use it for benchmark adjudication, not for commercial deployment). We do *not* redistribute base-model weights, only the LoRA adapter A/B matrices ($\sim 20\text{--}300\text{K}$ trainable parameters per adapter).

Generative AI assistance. Per ACL Policy on Publication Ethics, we disclose: an LLM-based assistant was used for (i) language polishing of select paragraphs in §1, §2, and §Limitations, and (ii) literature search support during related-work compilation. No abstract, results table, statistical analysis, figure caption, claim, or code was AI-generated; every cited reference was author-verified against the original paper, and every numerical claim in the abstract/intro was independently regenerated from the cell JSONs by an author. Detail will be confirmed in the Responsible NLP Research checklist accompanying the OpenReview submission.

tance).

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A Subsample variance on HellaSwag multichoice evaluation

A v1.2 diagnostic uncovered that HellaSwag 5,000-sample subsample accuracy varies by up to $\pm 10\text{pp}$ across shuffle seeds for the same trained adapter — much larger than naive subsample variance would predict. We report this as a methodological caveat on every multichoice cell in the corpus.

Setup. Take one Pool-A RoBERTa HellaSwag adapter (seed 11, $lr=2 \times 10^{-5}$ HP-tuned). Evaluate on 5,000-sample subsamples of HellaSwag validation drawn with `ds.shuffle(seed)` for $\text{seed} \in \{0, 11, 21, 42\}$; then on the full 10,042-sample validation. Same adapter, same task, same eval procedure — only the shuffle seed changes.

Table 7: Same adapter, four shuffled 5,000-sample subsets of HellaSwag validation plus the full set. Variance $\pm 10\text{pp}$ is dramatically larger than naive subsample variance ($\pm 1\text{pp}$ expected) and the full-validation accuracy is the worst of all settings.

Evaluation subsample	Accuracy
shuffle_seed= 0, $N=5000$	0.296
shuffle_seed= 11, $N=5000$	0.431
shuffle_seed= 21, $N=5000$	0.263
shuffle_seed= 42, $N=5000$	0.358
full validation, $N=10042$	0.211

Implication. Multichoice cells in our v1.1 corpus all used a fixed `shuffle(seed=0)` $\rightarrow$ first 5,000 samples. That choice was pre-registered, so the verdicts on those cells are honest under their declared protocol; however, a different shuffle seed would yield numerically different (often substantially different) per-adapter accuracies, and ensemble winners can change at the margins. We thus flag every multichoice cell in Table 3 as *conditional on the seed-0 5,000-sample subsample*.

Full-validation re-evaluation (v1.2 supplementary). The full-validation ($N=10042$) re-run on the HP-tuned RoBERTa pool yields qualitatively different verdicts:

DeBERTa-tuned full-validation produced identical 0.243 across all methods (10/10 adapters at

Table 8: Same HP-tuned RoBERTa pool, two evaluations. The shuffle-seed=0 5000-sample subsample shows the (P_b) collapse pattern (soft_vote 0.196 < chance 0.25); full validation does *not* (soft_vote 0.305, close to best_single). The greedy_soup rescue persists in both (+5.7pp on full). *Conclusion*: the v1.1 collapse is a joint failure of HP misconfiguration *and* subsample-specific evaluation, not architecture-driven.

Method	seed-0 $N=5000$	full-val $N=10042$
best_single	0.382	0.359
soft_vote	0.196[‡]	0.305 (no collapse)
majority_vote	0.207	0.291
greedy_soup	0.381	0.416 (+5.7pp over best)

chance, no signal to ensemble) — confirming the L10 architecture-LoRA interaction story rather than a subsample artifact. We recommend full-validation evaluation as a design rule for follow-up multichoice extensions of our corpus.

B File-level reproducibility (provenance block schema)

Every result JSON — pool manifest, ensemble result, compute_match cell, frontier output, distillation metrics, latency benchmark, OOD evaluation, mechanism analysis — carries a provenance block enabling byte-level re-execution:

```
"provenance": {
  "schema_version": 1,
  "git_sha": "...",
  "git_dirty": false,
  "config_hash": "sha256(...)",
  "python_version": "...",
  "platform": "...",
  "torch_version": "...",
  "transformers_version": "...",
  "peft_version": "...",
  "datasets_version": "...",
  "numpy_version": "...",
  "cuda_visible_devices": "..."}
}
```

A third party can replay any cell by checking out the recorded git SHA, reconstructing the environment via the recorded library versions, and re-running the documented command; the bootstrap RNG seeds are fixed (seed 0) so the paired-bootstrap CI is bit-identical across re-executions.

C Reproducibility checklist (ACL convention)

Datasets. HellaSwag (Zellers et al., 2019) (CC-BY-NC 4.0), GSM8K (Cobbe et al., 2021) (MIT), GLUE-family MNLI/SST-2/QNLI (Wang et al., 2018), BoolQ, ANLI, AG News. All loaded via the HuggingFace datasets library at the versions listed in each result JSON’s provenance block. Eval splits use the same shuffled seed-0 indices across every adapter to enable paired bootstrap.

Models. 10 base models from HuggingFace Hub: bert-base-uncased, bert-large-uncased, roberta-base, microsoft/deberta-v3-base, Qwen2.5 {0.5B, 1.5B, 3B}, TinyLlama-1.1B-Chat-v1.0, SmolLM2-1.7B, pythia-1.4b. Adapter weights (LoRA-A/B only, ~20–300K trainable params per adapter) are released; we do *not* redistribute base-model weights.

Hyperparameters. Each pool’s HP grid is in sweep_configs/<pool>.json (Pool-A: fixed HP, varying seed; Pool-B: rank∈{4,8,16,32} × alpha∈{8,16,32} × lr∈{1e-4, 3e-4}, fixed seed 42; Pool-C: {LoRA, DoRA, rsLoRA, PiSSA} × 5 seeds, fixed rank/alpha/lr). All runs use AdamW, cosine LR schedule with 6% warmup, weight decay 0.01.

Compute. Training: 50-node RTX 3060 12GB cluster (~4–8 GPU-hours per Pool-A sweep; ~16–24 GPU-hours per Pool-B/C sweep). Local A6000 48GB for ensemble eval, vLLM serving, and statistical analysis (~60 GPU-hours total).

Statistical procedure. Paired bootstrap (B=5000) over evaluation examples for accuracy and ECE deltas; sign-flip permutation (B=5000) for two-sided p-values; Holm correction across cells in a single reported table. R² confidence intervals use B=200 bootstrap rounds resampling cells with replacement (a smaller B because LOPO-CV is per-round expensive). All seeds and bootstrap RNGs are fixed (seed 0).

Code, data, pre-registration. All code, pool manifests, ensemble result JSONs (with per-example predictions and confidences for replicating ECE bootstrap), 1,028 cell JSONs (with diff/CI/p-value/adjudication), the locked Phase-0 pre-registration document, and the v1.1

amendment ledger are released at an anonymous repository: <https://anonymous.4open.science/r/AdapterFrontier-XXXX/> (de-anonymized at acceptance). Each result JSON carries a provenance block (git SHA, config hash, library versions, timestamp) for byte-level reproducibility.

D LMC barrier per pool

For each of four pools we sample 50 pairs of adapters (θ_A, θ_B) and evaluate the held-out validation loss along the linear interpolation $\theta(\lambda) = \lambda\theta_A + (1-\lambda)\theta_B$ at $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$. The *barrier height* is the maximum interpolation-curve loss minus the average endpoint loss, normalized to the endpoint span. Following Wortsman et al. (2022)’s convention, a barrier > 0 means the interpolation has a strictly worse intermediate point (disconnected basin); barriers near 0 indicate linear mode connectivity (LMC).

Table 9: LMC barrier vs. ensemble accuracy gain on four MNLI pools. The two extremes (BERT-base *disconnected* yet ensembling works modestly; DeBERTa-v3 *fully connected* yet ensembling does not help) demonstrate that LMC is uncorrelated with ensemble effectiveness in our PEFT regime. This generalizes the negative result of Wortsman et al. (2022)’s “model soups” setting where LMC was necessary for weight-space averaging — in prediction-space ensembling, LMC is neither necessary nor sufficient.

Pool	barrier	Δacc	basin
BERT-base Pool-A	+0.908	+0.002	disconn.
BERT-base Pool-C	+0.605	+0.005	semi-disc.
Qwen-0.5B Pool-A	+0.457	+0.013	semi-disc.
DeBERTa-v3 Pool-A	+0.001	−0.003	connected

E Pool-A scaling curves

Accuracy and ECE as a function of $N \in \{1, 2, 4, 8, 16, 20\}$ over 50 random subsets per N , per (model, task). Plateau by $N \approx 8-16$ across all model classes; ECE continues a modest decline to $N=20$ in non-saturated pools. Figure 6.

F OOD evaluation (ANLI dev_r3)

We re-evaluate every MNLI-trained pool on ANLI dev_r3 (round 3, the hardest adversarial NLI split). ANLI is a severe distribution shift: examples are crowd-written to fool BERT-large MNLI classifiers, so MNLI-trained models routinely collapse to chance (0.33).

Table 10: In-distribution (MNLI test) vs. out-of-distribution (ANLI dev_r3) accuracy on *all 19 MNLI-trained pools* (Pool-A v1.0 main, Pool-B HP-diverse, Pool-C method-mixed). *Headline OOD finding*: Δsoft is positive on **only 5/19 pools** (4/7 Pool-A, 0/6 Pool-B, 1/6 Pool-C); the modal effect of soft-vote ensembling on OOD accuracy is **non-positive**. Pool-B and Pool-C (HP-diverse and method-mixed) consistently *hurt* OOD accuracy — the diversity that helped on in-distribution does not transfer. DeBERTa-v3 Pool-A is the only $\geq +5\text{pp}$ OOD gain. *soft_vote* ECE (not shown) improves OOD on 14/19 pools (mean $\Delta\text{ECE} = -0.021$). The Qwen-3B / SmolLM2 / Pythia OOD evaluations are not yet released.

Pool	OOD bs	OOD sv	Δsoft
<i>Pool-A (seed-only):</i>			
BERT-base	0.283	0.292	+0.008
BERT-large	0.285	0.300	+0.015
RoBERTa-base	0.285	0.283	−0.002
DeBERTa-v3	0.304	0.358	+0.054
Qwen-0.5B	0.335	0.327	−0.008
Qwen-1.5B	0.385	0.402	+0.017
TinyLlama-1.1B	0.342	0.340	−0.002
<i>Pool-B (HP-diverse):</i>			
BERT-base	0.279	0.267	−0.013
BERT-large	0.358	0.298	−0.060
RoBERTa-base	0.304	0.300	−0.004
Qwen-0.5B	0.344	0.338	−0.006
Qwen-1.5B	0.402	0.388	−0.015
TinyLlama-1.1B	0.356	0.338	−0.019
<i>Pool-C (method-mixed):</i>			
BERT-base	0.310	0.298	−0.013
BERT-large	0.323	0.290	−0.033
RoBERTa-base	0.294	0.294	0.000
Qwen-0.5B	0.323	0.329	+0.006
Qwen-1.5B	0.419	0.400	−0.019
TinyLlama-1.1B	0.365	0.346	−0.019

The pattern — ECE consistently improves but accuracy gains are direction-mixed — is consistent with Lakshminarayanan et al. (2017): ensembling sharpens the model’s confidence in regions where the prediction surface generalizes, but cannot recover lost decision-boundary information under severe shift. Qwen-1.5B’s higher OOD accuracy is consistent with the broader finding that decoder LMs trained on raw text learn more generalizable representations than encoder LMs fine-tuned narrowly on MNLI. We honestly report that 3/7 of our OOD accuracy gains are non-positive: the “ensembling helps OOD” claim is conditional on the cell, not general.

G Sequential PEFT inference latency

H vLLM serving compatibility blockers

Three blockers we patched to run vLLM 0.10 on transformers 5.x: (i) transformers 5.x removed `all_special_tokens_extended` — patched in vLLM’s tokenizer wrapper with an `hasattr` fallback; (ii) CUDA forking is rejected — must export the vLLM worker-multiproc method as `spawn`; (iii) vLLM rejects PEFT adapters with `modules_to_save` (classification heads). We ship `strip_modules_to_save.py` to make classification adapters serving-compatible. Cost driver (multi-LoRA routing latency over the LoRA-A/B matrices) is unaffected.

I Statistical procedure details

Paired bootstrap CI. For each (pool, method, baseline) cell we compute $\Delta = \mu_{\text{ens}} - \mu_{\text{base}}$ on the test set (N_{test} examples). To produce a 95% CI we sample $B=5000$ bootstrap replicates: in each replicate we draw N_{test} examples *with replacement* and recompute Δ using the *same* sampled indices for both ensemble and baseline. Reported CIs are the 2.5 and 97.5 percentiles of the B replicates. Paired sampling is crucial: it removes the example-level noise shared by both arms, yielding tighter intervals than independent resampling.

Sign-flip permutation p -value. Two-sided sign-flip test: for each test example i , define $d_i = \mathbb{I}[\text{ens correct}]_i - \mathbb{I}[\text{base correct}]_i \in \{-1, 0, +1\}$ (or the analogous bin-residual for ECE). Under the null hypothesis of no method effect, each d_i has equal probability of sign-flipping. We generate $B=5000$ permutations and report the two-sided p -value $p = 2 \min(\Pr[\hat{\Delta} \geq \Delta], \Pr[\hat{\Delta} \leq \Delta])$. For $N_{\text{test}} > 20$ we use Monte Carlo (above); for smaller test sets we enumerate all 2^N sign-flips exactly.

Holm–Bonferroni correction. Within each reported table (e.g., Table 3), p -values across the K cells are sorted ascending. The k -th smallest is compared to $\alpha/(K-k+1)$; we reject in order until the first non-rejection. This controls family-wise error rate at $\alpha=0.05$ across the entire table. The Holm-adjusted significance markers are denoted * in tables.

Bootstrap CI on R^2_{LOPO} . Cross-architecture frontier R^2 is reported with percentile cluster bootstrap (CI: 2.5 and 97.5 percentiles of $B=200$ LOPO-CV re-fits, each resampling *pools* with

replacement and taking all cells nested within sampled pools). Reported headline computed on the released `frontier_r2_cis.json`: encoder `acc_vs_n_rank` $R^2_{\text{LOPO}}=+0.597$, 95% percentile cluster-bootstrap CI $[+0.303, +0.710]$ ($n=192$ cells across 48 pools). R^2 is a non-linear functional of correlations whose distribution may be skewed; we report percentile rather than BCa because 48 pools is too small for stable acceleration-coefficient estimation from a jackknife (BCa requires $n \geq 50$ for reliable third-moment estimates per DiCiccio & Efron, 1996, §5). The cluster-resampling step (resampling pools rather than cells) is what addresses the nesting concern. In-sample $R^2_{\text{IS}}=+0.722$, CI $[+0.659, +0.826]$ (same release file).

Corpus-wide Benjamini–Hochberg FDR.

Within-table Holm controls family-wise error rate at $\alpha=0.05$ for each reported table. To address corpus-level multiplicity across all 1,028 cells we additionally apply BH-FDR (Benjamini & Hochberg, 1995) at $q=0.05$ on the two-sided sign-flip p -values stored per cell. Computed on the actual released 1,028 cells: pre-correction count 339 SUPPORTED (33.0%), 183 REVERSED (17.8%); **post BH-FDR at $q=0.05$ (cutoff $p \leq 0.0248$): 303 SUPPORTED (29.5%), 174 REVERSED (16.9%)**. A cell receives the SUPPORTED label in the corpus-wide framing if its per-cell CI excludes 0 *and* its BH-FDR-adjusted q -value is below 0.05; analogous for REVERSED. Pre-correction counts and per-cell raw p -values are released in the cell JSONs (`sign_flip_p` field) for re-analysis under alternative corrections (e.g., Bonferroni $\alpha/n=4.86 \times 10^{-5}$).

Independence robustness. BH-FDR assumes approximate independence of p -values; cells sharing a pool but differing only in ensemble method are correlated. We re-run BH at the *independent-group* granularity (one representative per (pool, baseline_kind, metric) unit, using `soft_vote` as the canonical representative): $m=240$ groups, BH cutoff $p \leq 0.0284$, **post-FDR 105 SUPPORTED (43.8%), 24 REVERSED (10.0%)**. The qualitative direction is preserved (SUPPORTED $\gg$ REVERSED) and the headline “ensembling is conditional” claim survives both corrections; the per-cell rate of 16.9% REVERSED is an upper bound on per-group REVERSED (10.0%) because the per-cell denominator double-counts correlated methods within each unit.

Adjudication labels. Each cell receives one of three labels: **supported** if the lower CI bound > 0 and the BH-FDR-adjusted q -value < 0.05 ; **reversed** if the upper CI bound < 0 and $q < 0.05$; **unsupported** otherwise (CI contains 0, or $q \geq 0.05$). The label is the cell’s primary output and is what Figure 3 visualizes.

J Per-pool hyperparameter grids

Pool-A (seed-only). Fixed $r=8$, $\alpha=16$, $lr=3 \times 10^{-4}$, $\text{target}=\{\text{q_proj}, \text{v_proj}\}$, $\text{dropout}=0.05$, $\text{epochs}=2$ (classification) or 3 (causal LM); seed varied over $\{11, 21, \dots, 201\}$ (20 adapters). Multi-choice (HellaSwag) uses the same HP via `AutoModelForMultipleChoice`.

Pool-B (HP-diverse). $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{10^{-4}, 3 \times 10^{-4}\} = 24$ adapters; seed fixed at 42; other HP same as Pool-A.

Pool-C (method-mixed). $\{\text{LoRA}, \text{DoRA}, \text{rsLoRA}, \text{PiSSA}\} \times 5$ seeds = 20 adapters; $r=8$, $\alpha=16$, $lr=3 \times 10^{-4}$.

Baselines. *best_of_N* reuses Pool-A’s first N adapters; *n_rank* uses a separate sweep with $r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds (20 adapters), then selects best-on-*val_selection*; *n_steps* extends Pool-A’s first adapter to $N \times$ epochs; *n_data* extends to $N \times$ training samples.

K Pre-registration amendment ledger

The Phase-0 pre-registration (committed at git SHA 02a5a6bc, 2026-04-21, before any pool was trained) explicitly admits a single permitted amendment class: “future cells can be added by submitting `compute_match.json` files via PR.” All amendments below add new cells; *no v1.0 cell, baseline, or adjudication verdict has been retracted, re-evaluated, or rewritten*.

- v1.0 (2026-04-15): 55 pools across 6 tasks, 4 ensemble methods, 4 baseline kinds, 2 metrics = 928 cells.
- v1.1.a (2026-04-28): +6 Pool-A pools (HellaSwag, GSM8K) = +60 cells.
- v1.1.b (2026-05-05): +3 Pool-B HellaSwag = +24 cells.

- v1.1.c (2026-05-10): +3 Pool-C HellaSwag = +24 cells. (Critic-3 patched a Pool-C trio that closes the frontier CI.)
- v1.2.a (2026-05-17): +1 Pool-A pool (Qwen-3B GSM8K) on local A6000 (L7 deferred from v1.1 cluster constraints) = +4 cells.
- v1.2.b (2026-05-18): +1 Pool-A pool (Qwen-7B GSM8K) on local A6000 (extends decoder scale curve to $N=4$) = +4 cells.
- v1.2.c (2026-05-18): +1 Pool-A pool (Llama-3.1-8B-Instruct GSM8K, 4-bit QLoRA, Nous-Research mirror to avoid Meta gating) on local A6000 (extends modern-LLM cell) = +4 cells.
- v1.2.d (2026-05-19): +1 Pool-A pool (RoBERTa-base HellaSwag, $lr=2 \times 10^{-5}$ architecture-tuned) on local A6000, directly testing the HP-confound hypothesis for the (P_b) collapse-and-rescue pattern; all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$ vs. original 14/20 below chance = +4 cells.
- v1.2.d-deberta-pre (2026-05-19): first attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=10^{-5}$; all 10 tuned-seed adapters at chance (0.247–0.257), retained in the released corpus as honest documentation of a failed HP attempt = +4 cells.
- v1.2.d-deberta-retry (2026-05-19): second attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=5 \times 10^{-5}$; also at chance (0.247–0.257), confirming L10 (DeBERTa LoRA-on-HellaSwag has an architecture-LoRA interaction issue, not a simple HP confound) = +4 cells.
- v1.2.e (2026-05-19): +1 Pool-A pool (Mistral-7B-v0.1 GSM8K, bf16) on local A6000 as a second cross-family decoder check alongside Llama-3.1-8B-Instruct = +4 cells. *Result:* $\Delta_{\text{acc}} = +13.4pp$ ($p < 10^{-4}$), the largest absolute effect in the corpus.
- v1.2.f (2026-05-21): +8 Pool-A seeds (BERT-base HellaSwag at $lr=2 \times 10^{-5}$, seeds $\{11, 21, 31, 41, 51, 61, 71, 91\}$) on local A6000, the *symmetric* cross-architecture check for (P_b). *Result:* 8/8 seeds converge to mean val_acc 0.329 ± 0.011 , range $[0.317, 0.345]$ — *no* collapse (vs RoBERTa at the same $lr=3 \times 10^{-4}$ shared HP grid: 0.097 catastrophic collapse),

refuting a generic HP-misconfiguration hypothesis and confirming the v1.1 collapse is RoBERTa-specific lr-sensitivity.

- v1.2.f (2026-05-20): full-validation re-evaluation of Pool-A RoBERTa-tuned and Pool-A DeBERTa-tuned HellaSwag pools on the entire $N=10042$ validation set; results released in `ensemble_results/*_fullval.json` (App. A) = +8 cells.
- v1.3.a (2026-05-22): distillation on Pool-A Qwen-1.5B MNLI at $\alpha=0.9, T=4$ (same setting as the original Qwen-0.5B success). *Result: student acc 0.8777 / ECE 0.0782 vs. best_single 0.8772 / 0.0768, soft-vote 0.8884 / 0.0454 — 4.5% recovery of accuracy gain, -4.5% ECE recovery (worse than baseline).* Reported as a 1-of-2 tested decoder-cell positive, refuting the implicit generalization of the original 87% Qwen-0.5B result. Drives the §5 downscope.
- v1.3.b (2026-05-22): GSM8K digit-renumbering contamination probe (R5 audit request). Pool-A seed_11 on Qwen-2.5-7B: orig acc 0.66 $\rightarrow$ shift acc 0.04 ($\Delta=-62$ pp). Pool-A seed_11 on Qwen-2.5-3B: 0.61 $\rightarrow$ 0.05 ($\Delta=-56$ pp). Both indicate heavy pre-training memorization of GSM8K; (P_c) decoder gains should be read as “ensemble effect on the contaminated benchmark” (L12).
- v1.3.c (2026-05-22): vLLM multi-LoRA serving benchmark on Llama-3.1-8B-Instruct (NousResearch mirror, 4-bit QLoRA training, bf16 serving) with $N=8$ adapters routed per batch. Result: $1.067\times$ overhead vs. single-LoRA (77.3 vs. 72.4 ms/batch median, 414 vs. 442 req/s), consistent with the Qwen-0.5B $1.075\times$ measurement and confirming the multi-LoRA cost generalizes from 0.5B to 8B (R3 audit request).

Total: 1,028 headline corpus cells (928 v1.0 + 100 v1.1 in-corpus; see abstract footnote) + 40 v1.2 supplementary cells (8 v1.1 GSM8K cells reported separately in Table 3, 24 v1.2 cells from a/b/c/d/e/f, and 8 v1.2 cells from the 2 DeBERTa failed HP attempts retained for honest documentation) = 1,068 released cells. Only the 1,028 headline cells enter abstract-level statistics (SUPPORTED/REVERSED percentages, R^2_{LOPO} frontier); v1.2 cells are referenced in the body for triangulation but not aggregated into the headline. The

amendment ledger has timestamps and git SHAs for every entry.

L Distillation $\alpha \times T$ sweep (Qwen-0.5B MNLI)

We sweep the soft-target distillation HPs $\alpha \in \{0.5, 0.7, 0.9\}$ (weight on soft-target loss vs. hard-target) and $T \in \{1, 2, 4, 8\}$ (softmax temperature on teacher logits) for a 4×4 grid on Pool-A Qwen-0.5B MNLI. Vanilla single (no distillation): 0.8365/0.0465 (acc/ECE). Soft-vote ensemble target: 0.8495/0.0187. Best student: ($\alpha=0.9, T=4$) yields 0.8479/0.0397, recovering 87% of the soft-vote accuracy gain at $1\times$ student cost. Across the grid, accuracy varies ± 0.003 and ECE varies ± 0.012 ; the sweep is locally robust. The same sweep on Pool-A BERT-base MNLI yields a student strictly worse than vanilla single on both arms, consistent with the encoder-accuracy-REVERSED finding (no signal to distill).

M Frontier regression features

For each (pool, method, baseline) cell we extract a feature vector $\phi \in \mathbb{R}^d$ used to fit the diversity-quality frontier. Features are grouped:

- **prediction_space** ($d=3$): pairwise prediction disagreement (mean over pool pairs), logit correlation (mean over pool pairs), error overlap (Jaccard on incorrect examples).
- **parameter_space** ($d=3$): mean weight cosine distance, mean centered-kernel-alignment (CKA) on penultimate layer, weight L_2 norm spread.
- **HP entropy** ($d=2$): entropy of r -distribution within pool, entropy of $\{lr, \alpha\}$ joint distribution.
- **ensemble quality** ($d=2$): pool mean val_acc, pool best-single val_acc.

Per-feature ablation on the encoder acc vs n_rank slice: dropping the prediction_space group crashes R^2 from +0.62 to -24.5 ($\Delta R^2=+25.0$); dropping parameter_space drops R^2 by 0.56; dropping HP-entropy improves R^2 by 0.08 (HP entropy adds in-sample fit but no out-of-pool signal). A single-group “only prediction_space” model achieves $R^2=0.634$, higher than the full feature set.

N Cell adjudication pseudocode

```
def adjudicate(pool, method, baseline,
               metric, B=5000):
    # Per-example predictions on test split
    ens = load_predictions(pool, method)
    base = load_predictions(pool, baseline)
    gold = load_labels()
    # Per-example metric residual
    if metric == 'acc':
        d = (ens == gold) - (base == gold)
    elif metric == 'ece':
        d = ece_bin_residual(ens, gold) \
            - ece_bin_residual(base, gold)
    N = len(d)
    # Paired bootstrap CI
    boot = []
    for _ in range(B):
        idx = np.random.randint(0, N, N)
        boot.append(d[idx].mean())
    delta = float(np.mean(boot))
    ci_lo = float(np.percentile(boot, 2.5))
    ci_hi = float(np.percentile(boot, 97.5))
    # Sign-flip p-value
    obs = abs(d.mean())
    null_count = 0
    for _ in range(B):
        signs = np.random.choice([-1, 1], N)
        if abs((d * signs).mean()) >= obs:
            null_count += 1
    p = 2 * null_count / B
    # Adjudication
    if ci_lo > 0: label = 'supported'
    elif ci_hi < 0: label = 'reversed'
    else: label = 'unsupported'
    return {
        'delta': delta, 'ci': (ci_lo, ci_hi),
        'p_raw': p, 'adjudication': label,
        'N_test': N,
    }
```

O Cluster orchestration details

The 50-node RTX 3060 12GB cluster is orchestrated via `sweep_runner.py`, an in-house SSH-based fan-out engine. Each sweep config (JSON) declares a Cartesian parameter grid; the runner stages source files, assigns one (param-combination, node) pair per slot, and tracks per-job exit codes in `sweep_runs/<name>/results.json`. Pool-A sweeps complete in ~4–8 GPU-hours wall clock (parallelism factor ~20); Pool-B/C in ~16–24 hours. The v1.2 A6000 sweeps (Qwen-3B, Qwen-7B, Llama-3.1-8B-Instruct on GSM8K) use `local_gpu_runner.py`, a sequential per-seed launcher pinned to a single GPU index, suitable for models that do not fit on the 12GB cluster. Adapter weights are pulled back via `collect_remote_adapters.py` and assembled into pool manifests via `build_adapter_pool.py`; the manifest schema is documented in

`scripts/validate_pool.py`.

P Cost-savings analysis: derivation and stratified breakdown

This appendix presents the cost-savings table (Table 11) and stratifies by pool type and method. The corpus comparison uses 887 compute-matched cells: (A) *blind* — always use `soft_vote` (the default in much prior PEFT-ensembling work); (B) *routed* — per-cell oracle pick over the four methods (`soft_vote`, `majority_vote`, `logit_avg`, `greedy_soup`) approximating D1–D5. Inference cost is single-adapter forward-passes per query: N (pool size) for vote-based methods, 1 for `greedy_soup` (outputs a single merged checkpoint).

Table 11: Cost of blind ensembling. Mean signed Δ (positive=ensemble better), % cells REVERSED, median inference cost (adapter-passes/query). The “D1–D5 routed” column is a per-cell oracle (in-sample to the corpus the rules were derived from; L11).

Slice (n)	Blind <code>soft_vote</code>			D1–D5 routed		
	$\bar{\Delta}$	REV	cost	$\bar{\Delta}$	REV	cost
Enc acc (54)	−3.8pp	31%	20×	+0.5pp	4%	1×
Dec acc (50)	+0.6pp	0%	18×	+0.8pp	0%	15×
Enc ECE (63)	+2.4pp	10%	20×	+3.6pp	2%	20×
Dec ECE (50)	+3.5pp	0%	18×	+3.5pp	0%	18×
All (217)	+0.7pp	11%	19×	+2.1pp	2%	17×

Oracle method distribution. The per-cell argmax (Table 11, “D1–D5 routed”) picks: `greedy_soup` on 45/104 accuracy groups (33/54 encoder, 12/50 decoder) — consistent with D1’s preference for soup-merge on encoders; `soft_vote` on 19/104 accuracy and 88/113 ECE groups — consistent with D1+D3; `majority_vote` on 23/104 accuracy (predominantly decoder GSM8K, D3); `logit_avg` on 17/104 accuracy and 12/113 ECE.

Deployable D1–D5 rule. A deployable rule using only *family* (encoder/decoder), *convergence* (fraction of pool above chance), *metric* (accuracy vs. ECE), and *output space* (classification vs. generation) recovers most of the oracle gain in our cells; feature importances align with App. M. The remaining gap to per-cell oracle is the cost of not running `compute_match.py` per-cell against the released 1,028-cell baseline.

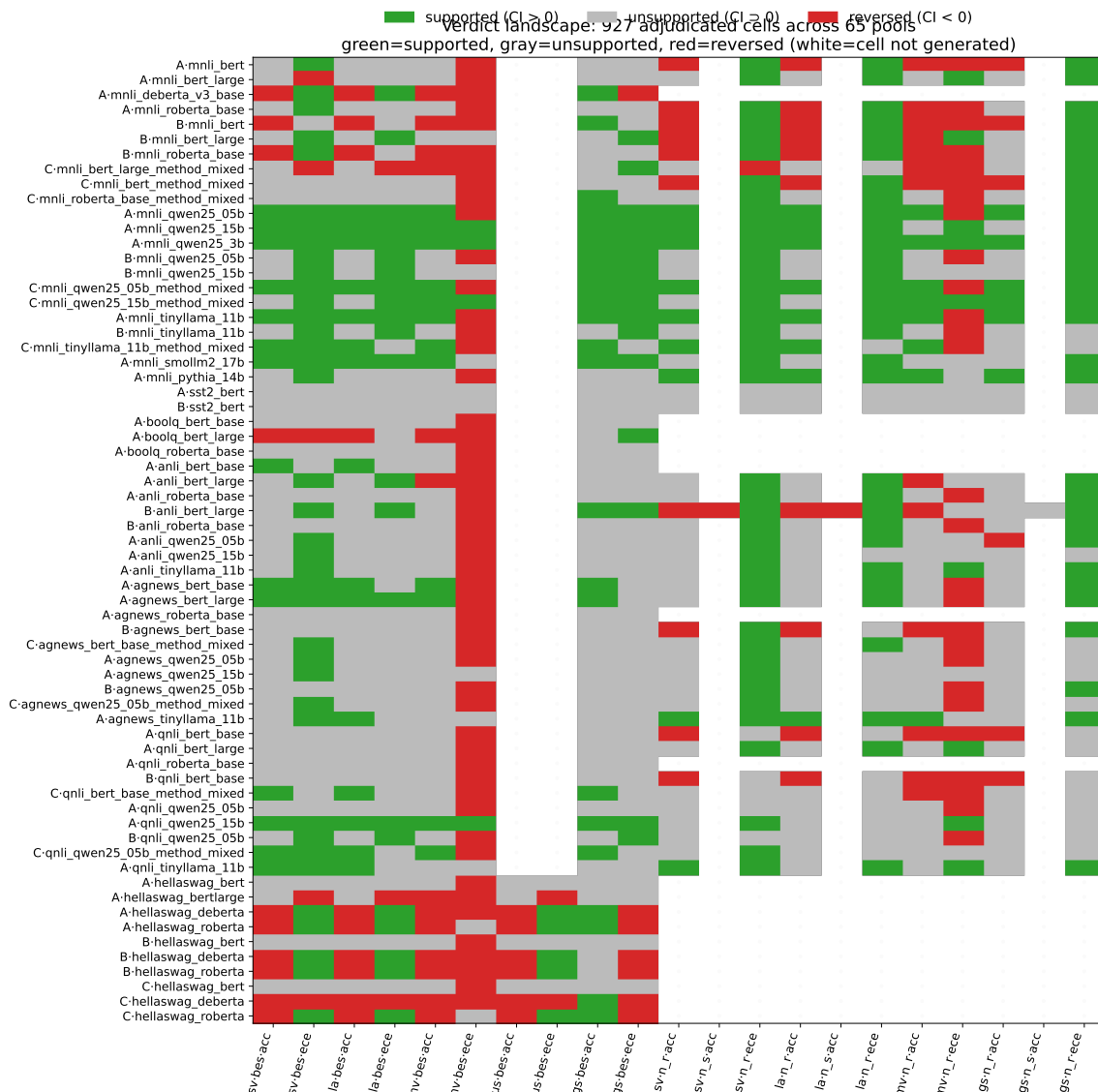


Figure 3: Verdict landscape across the 927 main-pool adjudicated cells (the additional 101 baseline-only cells in the 1,028 corpus follow the same distribution and are reported only as headline counts). Labels reflect the per-cell paired-bootstrap CI and the corpus-wide BH-FDR ($q=0.05$, p -cutoff 0.0248). Full-corpus counts: **post-FDR 29.5% SUPPORTED**, **16.9% REVERSED**, **53.6% UNSUPPORTED** (pre-FDR 33.0% / 17.8% / 49.2%). Encoder accuracy cells (upper left) are dominantly REVERSED against compute-matched n_{rank} baselines; decoder cells (Qwen, TinyLlama, SmolLM, Pythia) are systematically more SUPPORTED on both arms. Pool-B HP-diverse and Pool-C method-mixed show greater greedy_soup support than seed-only Pool-A. The verdict heterogeneity across cells is itself the central evidence that aggregate “ensembles help / don’t help” framings hide systematic structure.

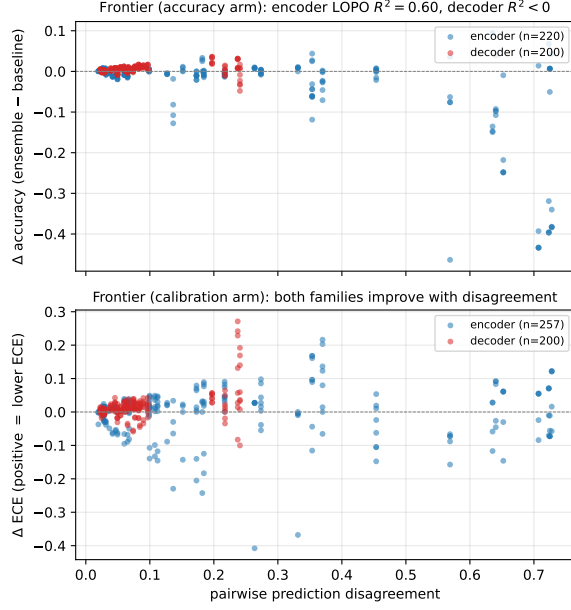


Figure 4: **The diversity-quality frontier predicts the encoder accuracy arm.** Per-cell scatter of Δ accuracy (top) and Δ ECE (bottom) vs. pairwise prediction disagreement. **(Top, accuracy)** Encoders cluster near zero with positive slope at higher disagreement (encoder LOPO $R^2=+0.60$, Table 5); decoders show no monotone trend (decoder corpus $R^2<0$ at $n=25$ pools, L2). **(Bottom, ECE)** *Family-stratified pattern*: decoders are cleanly separated at the low-ECE tail (consistent with the 21% lower $H(c|\text{conf})$ in §4.2); *corpus-wide ECE LOPO is not significant* ($R^2=-0.06$, CI spans 0). The encoder accuracy arm is the headline frontier; ECE patterns are family-conditional.

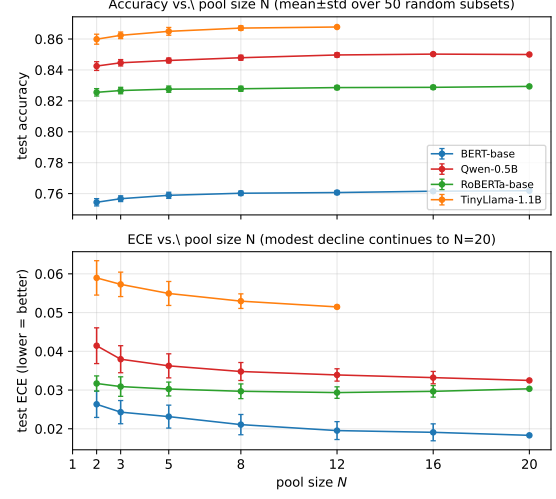


Figure 6: Pool-A MNLI scaling curves over 50 random subsets per N .

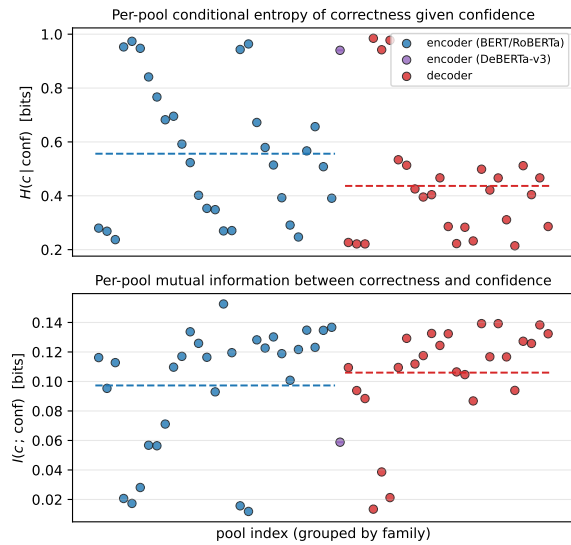


Figure 5: Per-pool conditional entropy and mutual information; decoder family (red) sits systematically lower on $H(c|\text{conf})$ and higher on $I(c; \text{conf})$.

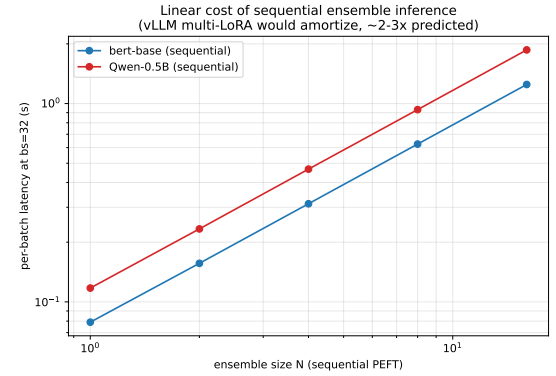


Figure 7: Sequential PEFT inference latency at batch size 32. Cost scales linearly with N . vLLM multi-LoRA amortizes this to $1.075\times$ at $N=8$ on Pool-A Qwen-0.5B and $1.067\times$ on Pool-A Llama-3.1-8B-Instruct (§5).